Lecture 2 - 25/10

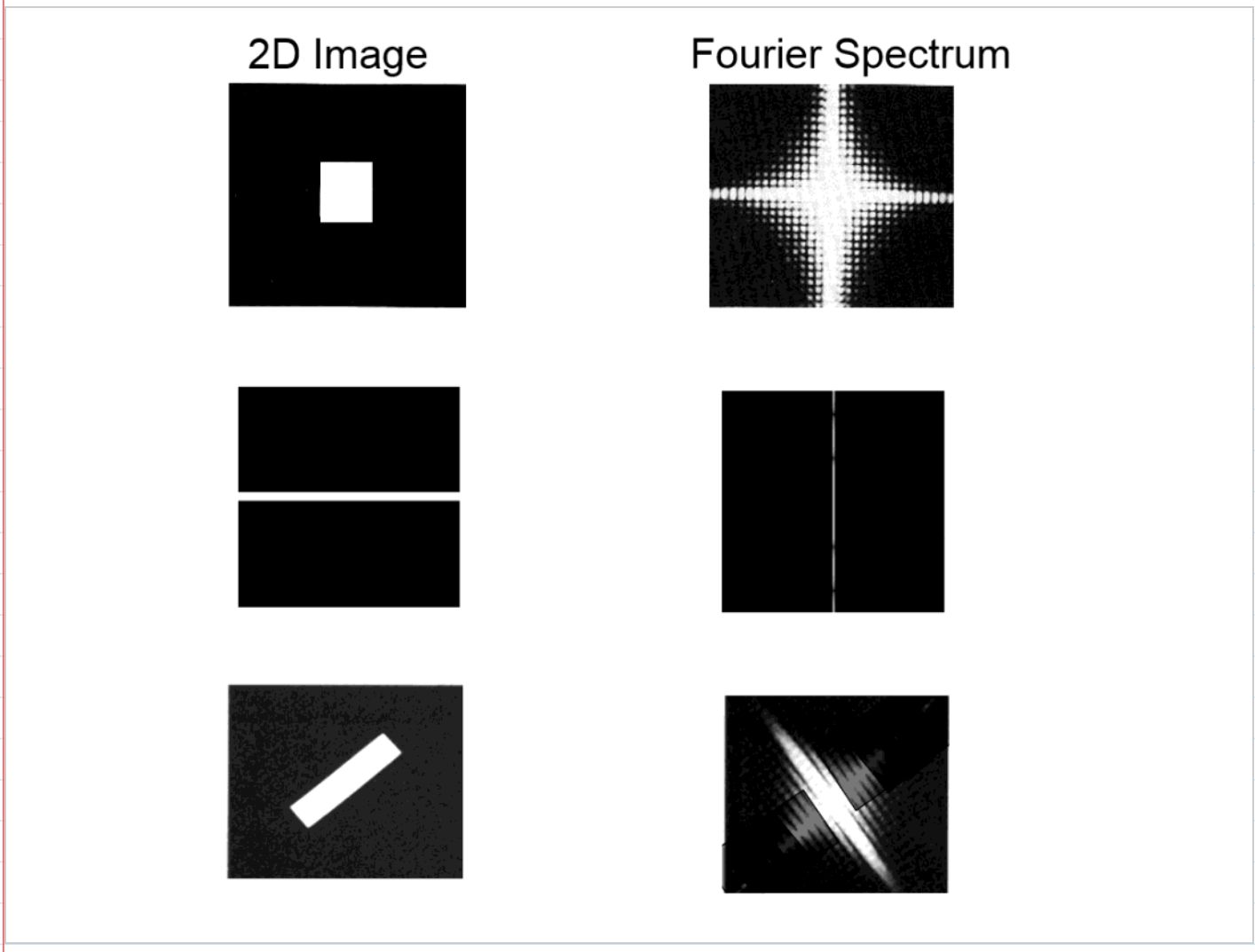
Friday, October 30, 2026 1:12 PM

FourierII_slides

FourierII_slides

Convolution_pre_ha...

https://docs.google.com/presentation/d/1YUv3qrtfDxR07KXgoe2oJHR3-aRCb2SiZr-yY_k8604/edit?slide=id.p1#slide=id.p1



Convolution

Flip g around x *and* shift it: $x \square 0$,
(*then* take the inner product)

1D (continuous, discrete):

$$f * g \quad (x) = \int_{-\infty}^{\infty} f(\alpha) g(x - \alpha) d\alpha$$

1D continuous

$$= \quad \text{D discrete}$$

Good for:
- **Pattern matching**
- **Filtering**
- **Understanding Fourier properties**

2D (continuous, discrete):

$$f * g \quad (x,y) = \int_{\alpha=-\infty}^{\infty} \int_{\beta=-\infty}^{\infty} f(\alpha,\beta) g(x-\alpha, y-\beta) d\alpha d\beta$$

2D continuous

$$= \sum_{}^{N-1} \sum_{}^{N-1} f(\alpha,\beta) g(x-\alpha, y-\beta)$$

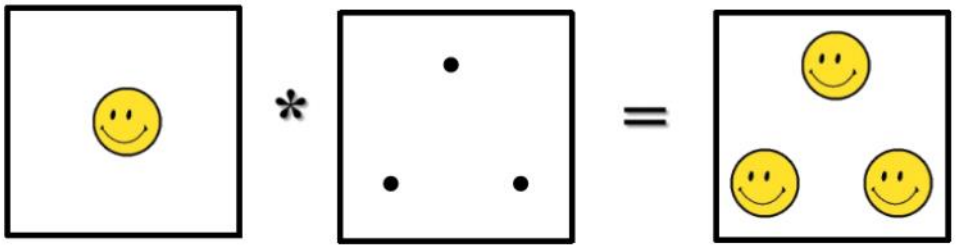
2D discrete

Examples

1) 

2) $f(x) * \delta(x) = f(x)$ **Unity**

3) $f(x) * \delta(x-d) = f(x-d)$ **Shift**

4)  **Replicate**

Convolution Properties

- Commutative:

$$f*g = g*f$$

- Associative:

$$(f*g)*h = f*(g*h)$$

- Homogeneous:

$$f*(\lambda g) = \lambda\, f*g$$

- Additive (Distributive):

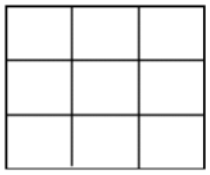
$$f*(g+h) = f*g + f*h$$

- Shift-Invariant

$$f*g(x-x0, y-yo) = (f*g)\,(x-x0, y-yo)$$

Proofs: Homework

Spatial Filtering Operations



Example 3x3 filter

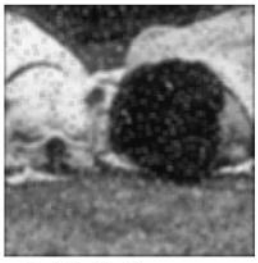
$$h(x,y) = f(x,y) * \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$h(x,y) = 1/9 \sum_{(n,m)} f(n,m)$$

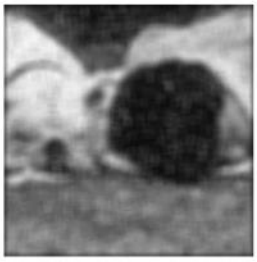
Average of all pixels in the 3x3 neighborhood of (x,y)



Salt & Pepper Noise

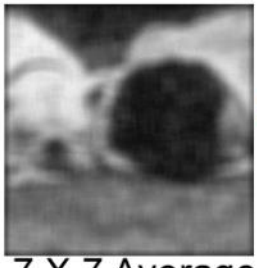


3 X 3 Average



5 X 5 Average

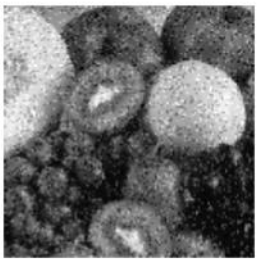
Local averaging:
Removes noise but blurs edges



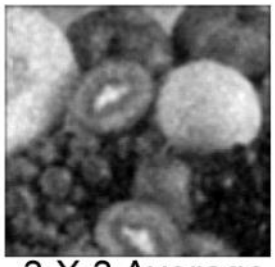
7 X 7 Average



Median



Salt & Pepper Noise



3 X 3 Average

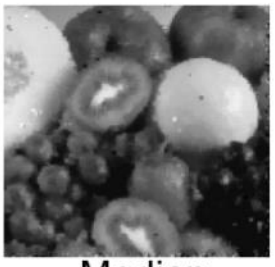


5 X 5 Average

Local averaging:
Removes noise but blurs edges



7 X 7 Average



Median



$$f(x,y) * \begin{bmatrix} 1 & -1 \end{bmatrix}$$

$$= f(x+1,y) - f(x,y)$$

x derivative $\frac{\partial f}{\partial x}$

$$f(x,y) * \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

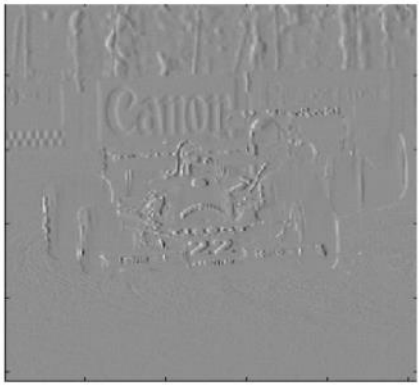
$$= f(x,y+1) - f(x,y)$$

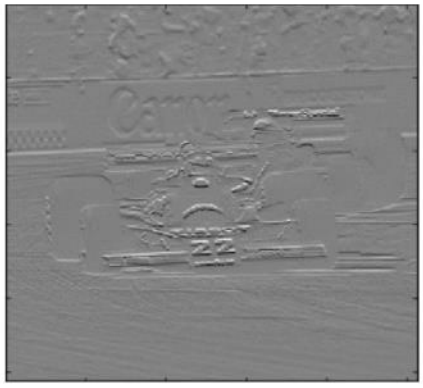
y derivative $\frac{\partial f}{\partial y}$

A very simplistic "Edge Detector"

Gradient magnitude

$$\|\nabla f\| = \sqrt{f_x^{\,2} + f_y^{\,2}}$$







The Convolution Theorem

$$\Im\{f(x) * g(x)\} = \Im\{f(x)\} \cdot \Im\{g(x)\}$$

Proof: Homework

and similarly:

$$\Im\{f(x)g(x)\} = \Im\{f(x)\} * \Im\{g(x)\}$$

➔ Computational complexity of Discrete Convolution:

O(N *log*N)

Take f and g → FFT → $F \cdot G$ → FFT^{-1} → $f * g$

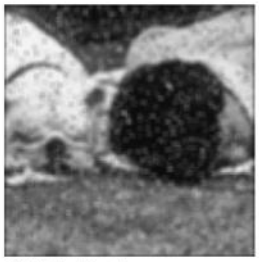
Going back to the "noise cleaning" example...



Salt & Pepper Noise

$$* \quad \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad = \quad$$

rect

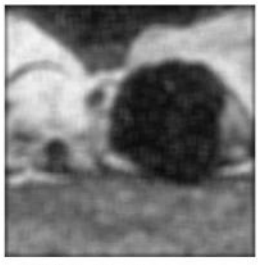


3 X 3 Average

Convolution with a ***rect*** □□ Multiplication with a ***sinc*** in the Fourier domain

= ***LPF*** (Low-Pass Filter)

□ **blurs the image**



5 X 5 Average

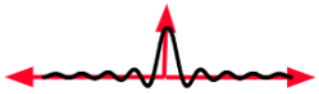
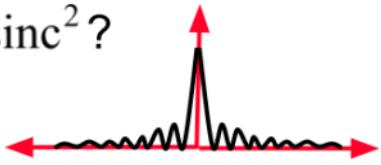


7 X 7 Average

Wider ***rect*** □□ Narrower ***sinc*** = Stronger ***LPF*** □ **Stronger blur**

Examples

What is the Fourier Transform of $f(x) = \text{sinc}^2$?

$f(x) = ($  $)^2 =$ 

$$\Im\{f(x)\} = \Im\{\text{sinc} \cdot \text{sinc}\}$$

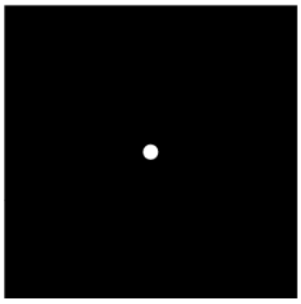
$$= \Im\{\text{sinc}\} * \Im\{\text{sinc}\}$$

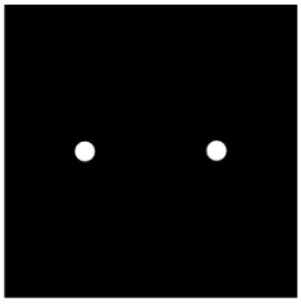
$$= rect * rect$$

$=$ 

$=$ 

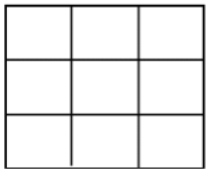
Image Domain





LPF in *Image domain* vs. *Fourier domain*

Computational complexity of convolution: O(N*log*N) via Fourier.
But for filter with a small support k □ O(kN) in the Image domain.

1) $f(x,y) *$ 

$\leftrightarrow$ $F(u,v)\ sinc(u,v)$

Convenient
(small spatial support;
no cyclic assumption)

But not an ideal LPF...

2) $f(x,y) * sinc(x,y)$ $\leftrightarrow$ $F(u,v)\ rect(u,v)$

An ideal LPF

3) $f(x,y) * gaussian(x,y) \leftrightarrow F(u,v)\ Gaussian\ (u,v)$

The Sampling Theorem

Developed on the board (also found in recording):

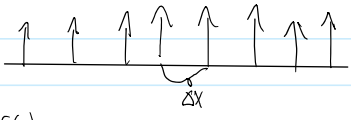
- **Nyquist frequency**
- **Aliasing**
- **Ideal interpolation**
- **Discretization induces a Cyclic function**



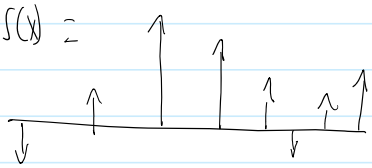
function



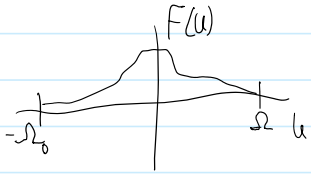
sampling func $S(x) =$

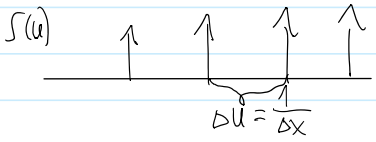


$f(x) \cdot S(x) =$

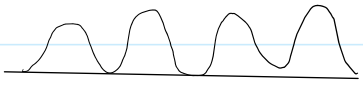


Fourier





$f(u) * S(u) =$



lets look at cases

if F(u) is band limited -

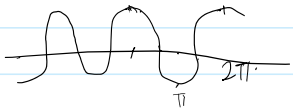
there is a certain frequency, above which all frequencies are 0, then we can sample it discretly

$\exists\, \Omega_0$ s.t. $\forall |u| > \Omega_0, F(u) = 0$

convolution with many dirac duplicates

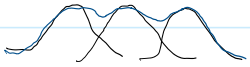
the "Nyquist frequency"

case 1: no overlap, if $\Delta u > 2\Omega_0 \Rightarrow \Delta x < \frac{1}{2\Omega_0}$



from here, take a LPF that captures one of the duplicated, thusly the original F(w), then go to f(x)
so if we sample more dearly than $2\Omega_0$, we can reconstruct

Case 2: overlap



how if we take a LPF of this we get 
this causes aliasing

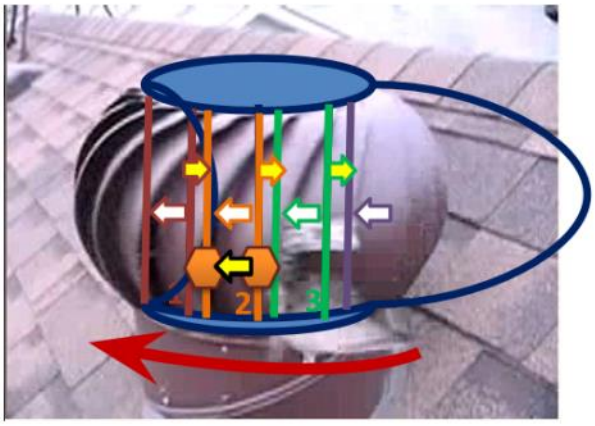
Examples of Temporal Aliasing

A "fast" dynamic event imaged by a "slow camera"

Motion aliasing



Examples of Temporal Aliasing



What's going on…?!

- The turbine rotates fast to the left.
- Each blade moves almost to the next blade position between frames.
  - So, we wrongly perceive rotation of the blades to the right
    - ***TEMPORAL ALIASING!***

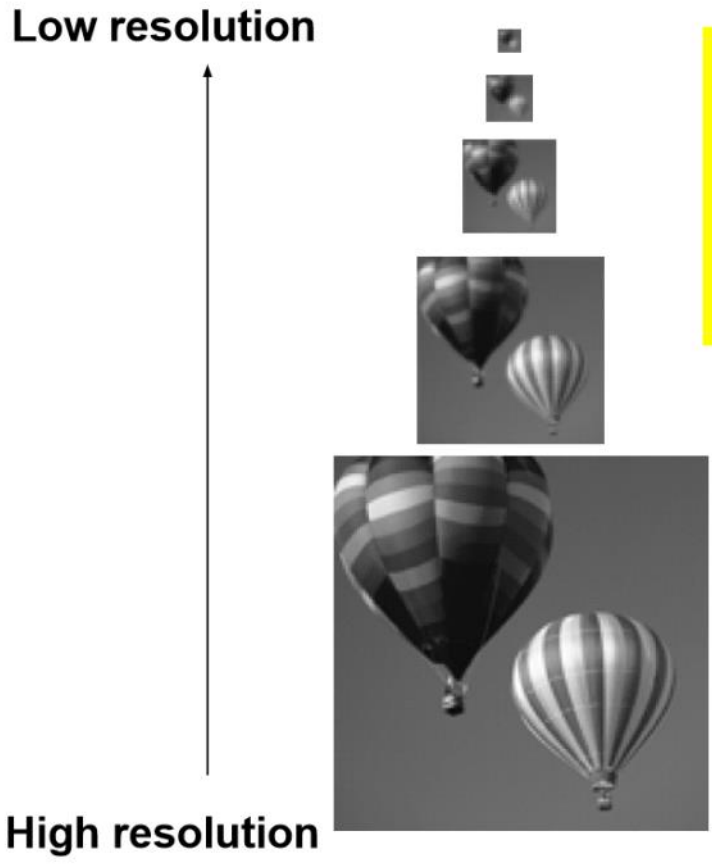
It happens because the temporal sampling rate (framerate) is too slow.

Multi-Scale Image Representation

- Gaussian pyramids
- Laplacian Pyramids
- Wavelet Pyramids



Image Pyramid



Good for:

- pattern matching
- motion analysis
- image compression
- other applications

Fast Pattern Matching

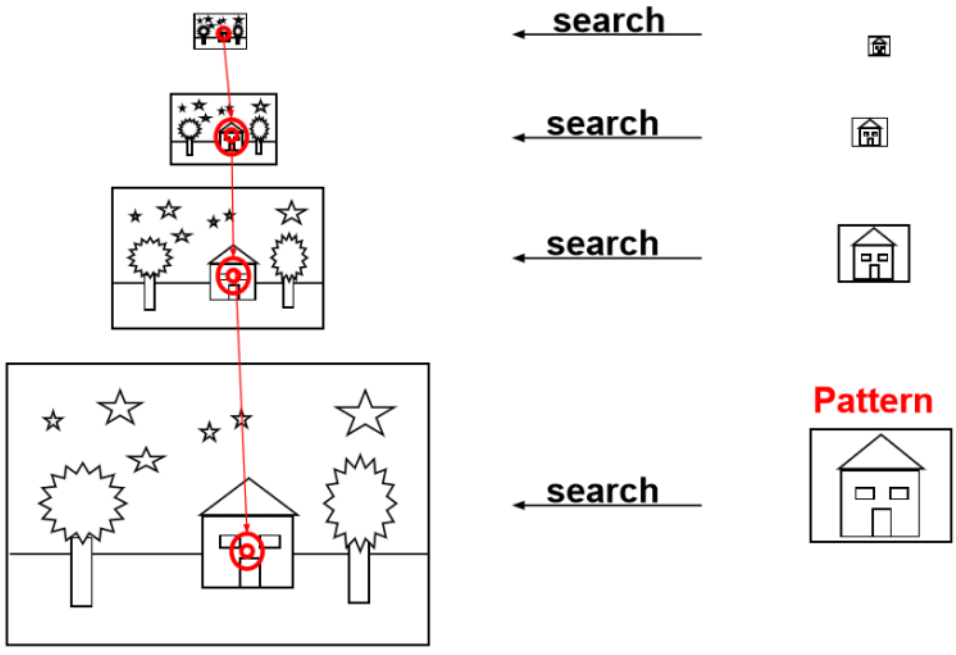
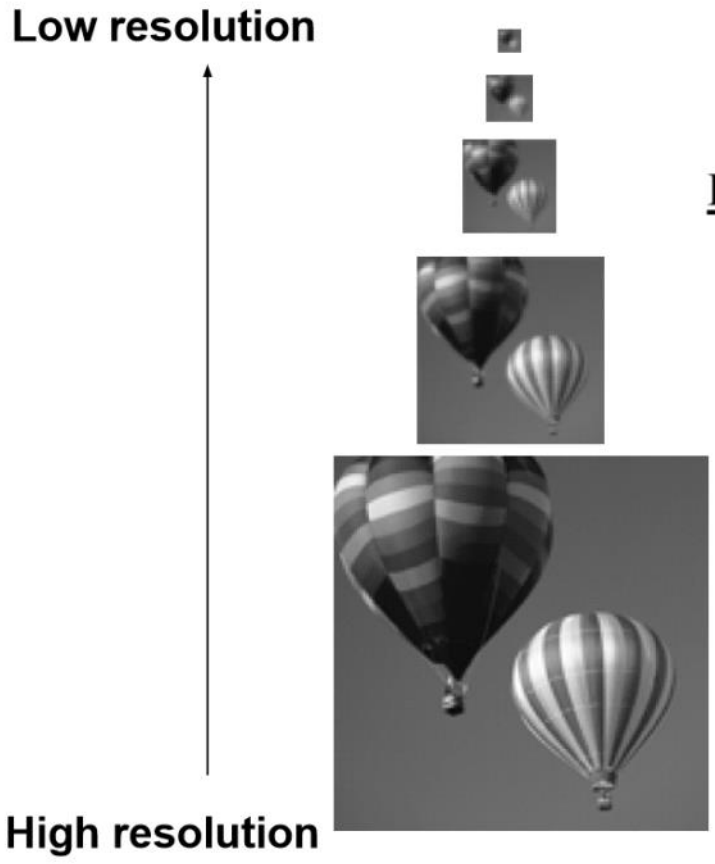


Image Pyramid

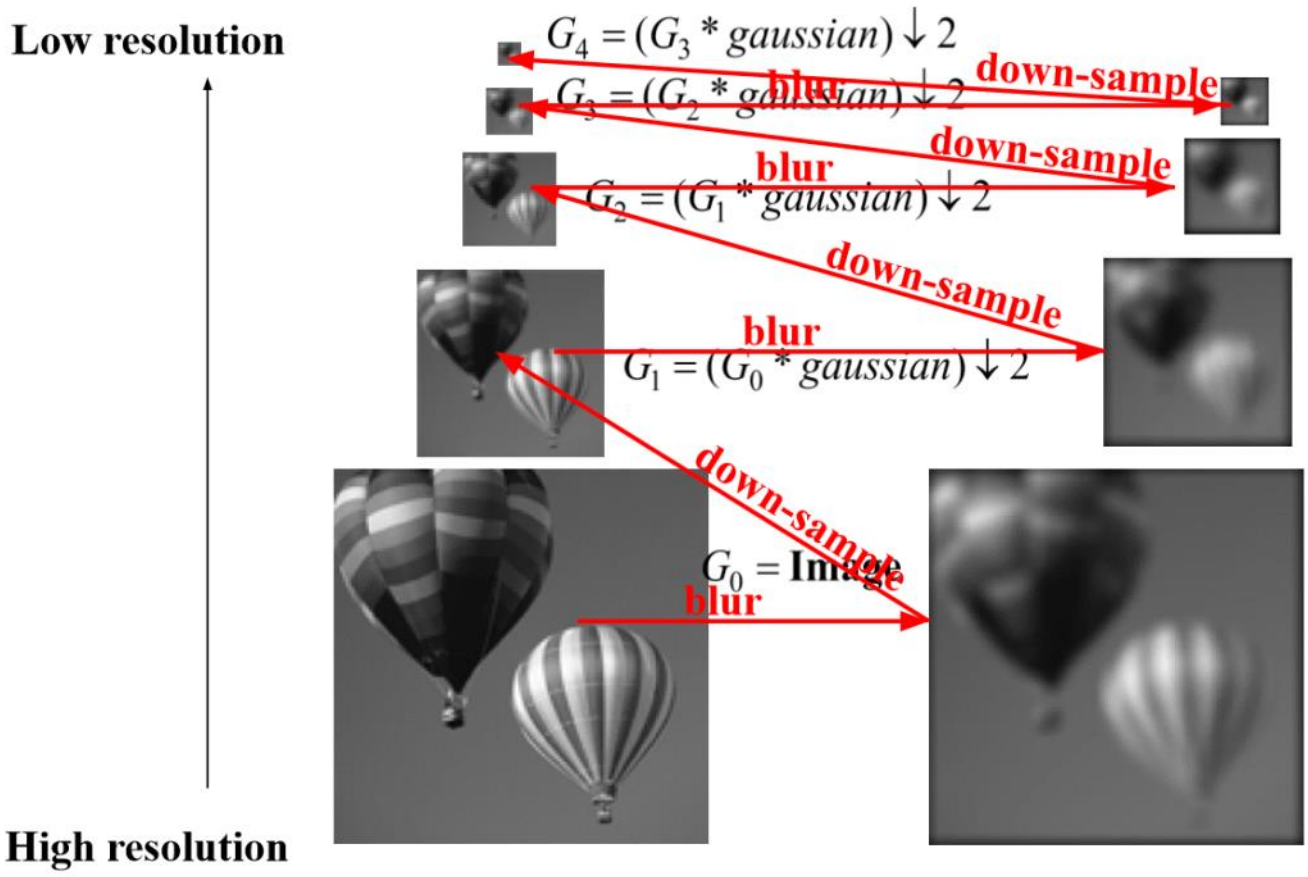


Does not consume much space

For an NxM image:

$$NM + \frac{1}{4}NM + \frac{1}{16}NM + \ldots$$
$$= \sum_l (\frac{1}{4})^l NM$$
$$\leq 1\frac{1}{3} NM$$

The Gaussian Pyramid



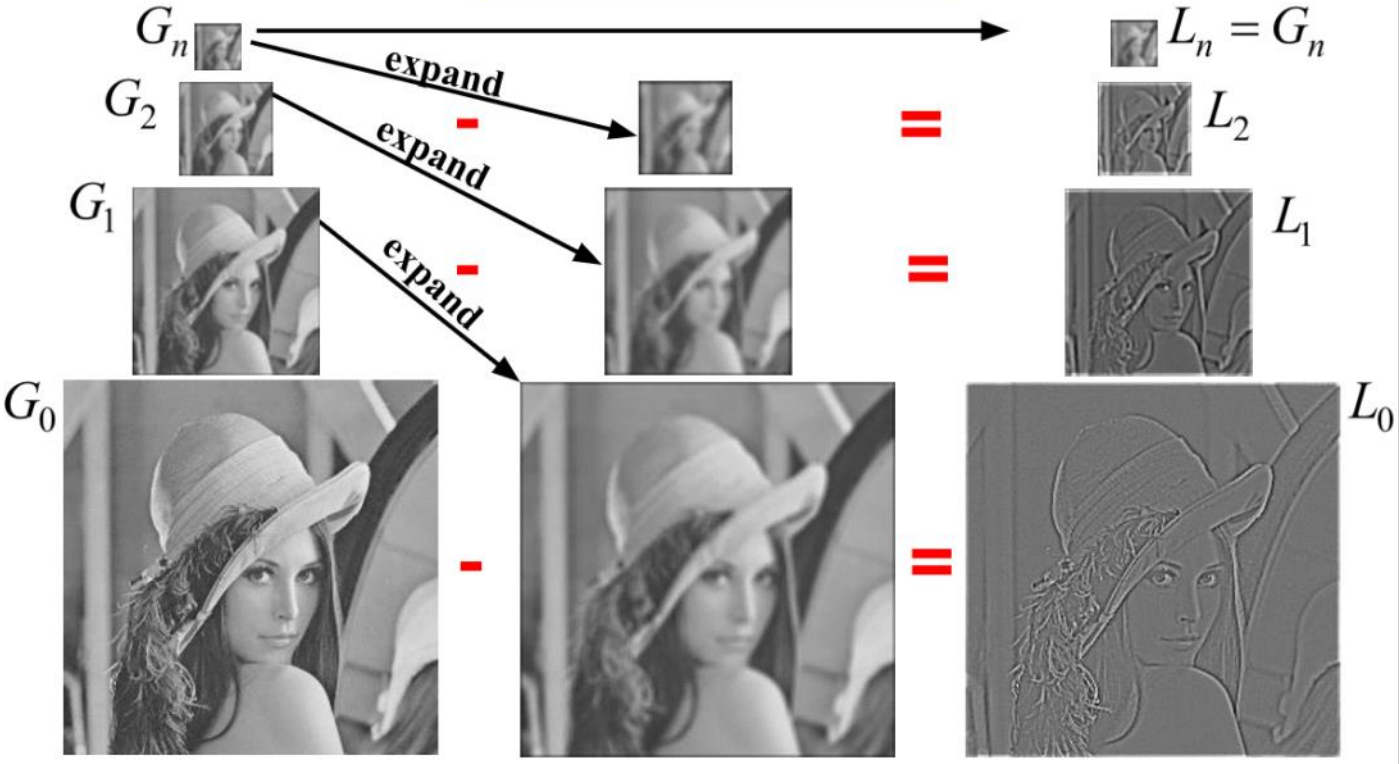
The Laplacian Pyramid

$$L_i = G_i - \text{expand}(G_{i+1})$$

$$G_i = L_i + \text{expand}(G_{i+1})$$

Useful for image compression

Gaussian Pyramid **Laplacian Pyramid**



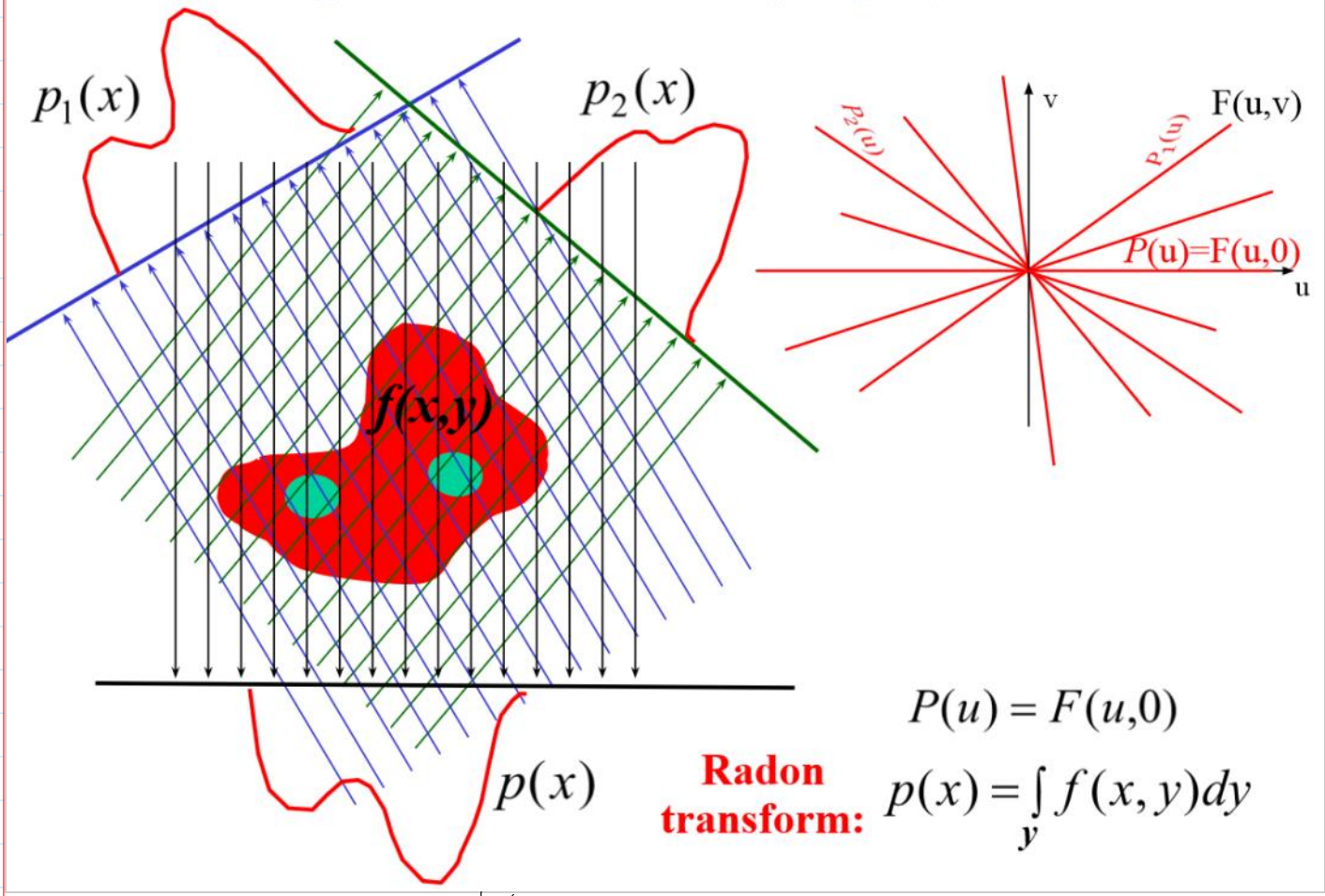
Laplacian ~ Difference of Gaussians



DOG = Difference of Gaussians

More details on Gaussian and Laplacian pyramids can be found in the paper by Burt and Adelson (link will appear on the website).

Computerized Tomography (CT)



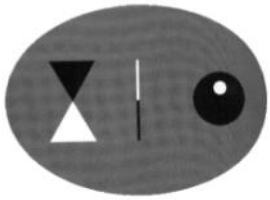
$$P(u) = F(u,0)$$

Radon transform: $$p(x) = \int_y f(x,y)dy$$

$$P(u) = \int_x p(x)\, e^{-2\pi i u x}\, dx = \int_x \int_y f(x,y)\, dy\, e^{-2\pi i u x}\, dx =$$

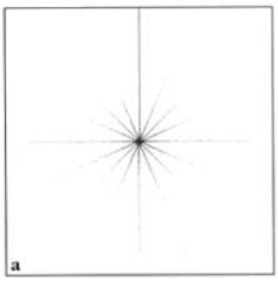
$$= \iint_{x\,y} f(x,y)\, e^{-2\pi i (ux + 0y)}\, dy\, dx = F(u,0)$$

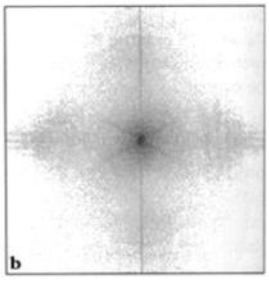
Computerized Tomography (CT)



Original (simulated) 2D image

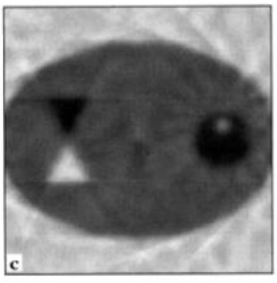
8 projections- Frequency Domain

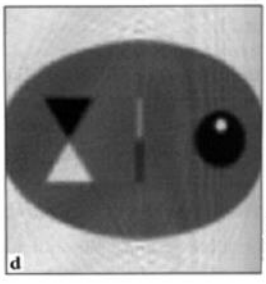




120 projections- Frequency Domain

Reconstruction from 8 projections





Reconstruction from 120 projections

End of Lesson...

Exercise#1 -- will be posted on the website.
(Theoretical exercise: To be done and submitted individually)

To get course announcements:
please sign in to the course website! (Moodle)

Vision & AI Seminar (not for credit):
Thursdays at 12:15-13:15
(in person – in Ziskind-1)